A map whose fibers are contractible can be made into a half adjoint equivalence

The Univalent Foundations Program

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Editorial note. The counted statement is the author thm environment at source lines 447-449 and its proof environment opens on line 450, the next line; the pairing is by adjacency and the retained proof range 450-458 is the whole environment.

Editorial note. The retained counter values (chapter 4, section 4, theorem offset 2) make the counted theorem print as Theorem 4.4.3, matching the original book numbering read from the label-number dump of the pinned snapshot (newlabel thm:lequiv-contr-hae = 4.4.3).

Editorial note. One label the proof cites lies outside the card, lem:coh-hfib (book Lemma 4.2.11, defined at line 340 of this chapter pinned file), and is declared in external_references as a non-load-bearing navigation pointer: it supplies the reduction of the coherence datum to a path in a fiber.

Editorial note. The retained preamble is the book own macro library and layout adapter as stored for this chapter; no package outside it is requested.

Editorial scope. One original result of the pinned HoTT Book source tree, counted once: contractible fibers give a half adjoint equivalence (author label thm:lequiv-contr-hae, book Theorem 4.4.3).

Allowed standard prerequisites (editorial).

- Contractibility of the fibers. The hypothesis gives, for every point y of the codomain, a center and a contraction of the fiber of f over y.. It is the structure the construction reads off.
- Half adjoint equivalence data. The data of a half adjoint equivalence are an inverse map, a homotopy from the composite to the identity, and a coherence path relating the two triangle identities.. It is the structure that has to be produced.
- Coherence in the fiber. Giving the coherence datum is the same as giving, for each x, a path in the fiber of f over f(x) between two specified points.. It is the reduction that makes the construction possible.

Source reference. Original lemma 4.2.11: sources/hott-equivalences/pinned-original-source.tex, source lines 340–340. Book Lemma 4.2.11, identifying the coherence datum of a half adjoint equivalence with a path in a fiber, cited at

source line 456. Navigation pointer only: the retained proof builds the path it asks for.

Theorem 4.4.3. *For any* $f : A \to B$ *we have* $\mathsf{isContr}(f) \to \mathsf{ishae}(f)$.

Proof. Let $P : \mathsf{isContr}(f)$. We define an inverse mapping $g : B \to A$ by sending each $y : B$ to the center of contraction of the fiber at y:

$$g(y) :\equiv \mathsf{pr}_1(\mathsf{pr}_1(Py)).$$

We can thus define the homotopy ϵ by mapping y to the witness that $g(y)$ indeed belongs to the fiber at y:

$$\epsilon(y) :\equiv \mathsf{pr}_2(\mathsf{pr}_1(Py)).$$

It remains to define η and τ. This of course amounts to giving an element of $\mathsf{rcoh}_f(g, \epsilon)$. By original lemma 4.2.11, this is the same as giving for each $x : A$ a path from $(gfx, \epsilon(fx))$ to (x, refl_{fx}) in the fiber of f over fx. But this is easy: for any $x : A$, the type $\mathsf{fib}_f(fx)$ is contractible by assumption, hence such a path must exist. We can construct it explicitly as

$$\big(\mathsf{pr}_2(P(fx))(gfx, \epsilon(fx))\big)^{-1} \cdot \big(\mathsf{pr}_2(P(fx))(x, \mathsf{refl}_{fx})\big). \qquad \square$$